in the target point and with radius equal to the minimum distance found so far: and it is highly likely that this sphere will intersect many of the branches' hyper-cubes in at least one dimension.

To overcome these limitations, we could try a few approaches:

- We could use different criteria to decide where to partition the k-dimensional space, using different heuristics:
  - Don't use splitting line passing through points, but just divide a region in two balanced halves (either with respect to the number of points or the sub-regions size: basically, choose the mean instead of the median);
  - Instead of cycling through dimensions, choose at every step the dimension with the greatest spread or variance, and store the choice in the tree node.
- Instead of storing points in nodes, each node could describe a region of space and link (directly or indirectly) to an array containing the actual elements.
- We could approximate nearest neighbor search: for instance, we could use *locality sensitive hashing*.
- Or, we could find new ways to partition the k-dimensional space, ideally trying to reduce sparsity.

Heuristics help on some datasets, but in general won't solve the issue with higher dimensional spaces.